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REVIEW/SURVEY PAPERS

A Comprehensive Survey on Deep Learning Techniques in Educational Data Mining

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Abstract

Educational Data Mining (EDM) has emerged as a vital field of research, which harnesses the power of computational techniques to analyze educational data. With the increasing complexity and diversity of academic data, Deep Learning techniques have shown significant advantages in addressing the challenges associated with analyzing and modeling this data. Existing studies are scattered across various domains, making it challenging to gain a comprehensive understanding of how Deep Learning techniques can transform educational practices. This survey aims to systematically review the state-of-the-art in EDM with Deep Learning. We begin by providing a brief introduction to EDM and Deep Learning, highlighting their relevance in the context of modern education. Next, we present a detailed review of Deep Learning techniques applied in four typical educational scenarios, including knowledge tracing, student behavior detection, performance prediction, and personalized recommendation. Furthermore, a comprehensive overview of public datasets and processing tools for EDM is provided. Finally, we point out emerging trends and future directions, aiming to guide researchers and practitioners in advancing the field of EDM.

Keywords Educational data mining · Deep learning · Reinforcement learning · Educational datasets

1 Introduction

Deep Learning has experienced remarkable advancements in recent years, revolutionizing diverse domains, including education. Deep Learning, a form of machine learning, relies on artificial neural networks to facilitate the discovery of hierarchical features, which enhances the ability to recognize patterns [1]. In contrast to conventional machine learning approaches that require manual feature engineering, Deep Learning allows a machine to automatically discover intricate structures in large data by using multiple layers of abstraction. This layered feature learning process enables Deep Learning models to learn complicated patterns in data and achieve state-of-the-art performance across domains such as speech recognition [2], image classification [3], and natural language processing [4].

In general, three primary categories of algorithms exist in the domain of Deep Learning [5]: supervised learning, which models learn from labeled data for tasks such as image classification using Convolutional Neural Networks (CNNs) and sequence prediction using Recurrent Neural Networks (RNNs) or Transformers [6]; unsupervised learning, which aims to discover patterns in unlabeled

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data, enabling tasks like data generation with Generative Adversarial Network (GAN) [7] and feature learning with Autoencoders [8]; and reinforcement learning, in which an agent learns optimal behavior through interaction with an environment. The goal of the agent is to optimize the cumulative reward. Applying Deep Learning algorithms suitable for different application scenarios [9, 10] can greatly increase their performance. With increasing computing power, Deep Learning has achieved major breakthroughs and outstanding results in many fields.

Education fields have been revolutionized by traditional machine learning and Deep Learning algorithms, and exploring previous research is key to understanding its applications. There is a main term that is currently used in education, Educational Data Mining (EDM) [11]. This field is typically interdisciplinary, including but not limited to information retrieval, data analysis, psychopedagogy, cognitive psychology, etc. EDM encompasses computerized techniques and tools that facilitate the automated identification and extraction of meaningful patterns and valuable information from extensive datasets obtained within educational settings [12–14]. While this involves pattern extraction similar to general data mining, EDM's distinctiveness is critical for understanding its scope and significance. Unlike general approaches applicable to any domain, EDM operates exclusively within educational contexts, analyzing unique data types intrinsically linked to learning activities (*e.g.*, student interaction logs, assessment results, forum discussions). More importantly, its goals are fundamentally pedagogical: aiming not merely to discover patterns but to generate actionable insights that directly inform teaching practices, personalize student learning pathways, identify learners needing support, and ultimately enhance educational effectiveness. This inherent focus on improving learning and teaching, often grounded in educational theory, defines EDM's unique contribution and relevance [15, 16].

In fact, researchers have also utilized diverse traditional machine learning algorithms [17] for EDM across varied educational contexts [18]. However, traditional machine learning algorithms come with certain limitations and practical challenges in EDM [19]. First, data quality and integration are major issues. Educational data often comes from multiple, inconsistent sources and contains missing or noisy entries, making it difficult to build reliable models. Second, evaluation and validation are complex because learning outcomes are multifaceted and can not be easily captured by single metrics, leading to potential overfitting or misinterpretation of results. Third, fairness and privacy concerns arise as educational data frequently includes sensitive student information. Traditional algorithms do not inherently account for biases or ensure data protection, posing ethical and legal risks.

Integrating Deep Learning into educational scenarios is driven by the aspiration to harness the potential of Artificial Intelligence and machine learning, thereby enriching the teaching and learning experience [1, 20]. Deep Learning models possess remarkable capabilities, enabling them to effectively process and analyze vast amounts of educational data. By uncovering meaningful patterns and making accurate predictions, these models provide valuable insights that can inform and enhance educational practices. Educators and researchers can leverage these insights to adapt teaching strategies, personalize instruction, and optimize learning outcomes. By harnessing the power of Deep Learning, the educational community can unlock new opportunities for improved efficiency, effectiveness, and adaptability in education through intelligent data processing and adaptive methodologies [21].

Deep Learning has also made significant achievements in completing specific scenario tasks in EDM. Currently, EDM scenarios are often divided into: knowledge tracing, student behavior detection, performance prediction, and personalized recommendation. Each subfield has distinct specific data input patterns and task requirements. Deep Learning-based knowledge tracing algorithms [22] can be classified into Deep Knowledge Tracking (DKT) and its variants, *e.g.*, DKT based on Memory networks, Attention mechanisms [23] and Graph structures [24]. Complex neural network models can be applied to Student Dropout Prediction (SDP), which is a task of student behavior detection [25]. Similarly, neural networks can also provide quite reliable accuracy in the EDM scenario of performance prediction [26]. In the field of personalized recommendation [27], hybrid techniques-based recommendation algorithms may be gradually taking the dominant position, but the emerging privacy issues also need close attention [28].

Deep Learning techniques offer numerous advantages over traditional machine learning methods when applied in educational scenarios [29, 30]. One notable advantage is the automatic learning of hierarchical representations from raw data, eliminating the need for manual feature engineering. This characteristic makes Deep Learning models highly suitable for analyzing various types of educational data. Deep meta-learning [31] can also be employed to assess and compare the performance of different algorithms on EDM tasks. In addition, we can explore the implementation and optimization of federated learning algorithms [32] in EDM. It enables model training on local data without having to share it with a central server, thereby protecting student information. A telling difference is shown by research [20] that approximately 67% of papers report Deep Learning demonstrated superior performance compared to the traditional machine learning baselines in all conducted experiments.

Data Collection Methodology. Followings are rules we applied to include or exclude papers:

- Search terms: Deep Learning and Educational Data Mining are the two keywords mainly involved in our survey. In order to access more related publications, we also used specific educational scenarios as our search terms (*e.g.*, knowledge tracing, performance prediction etc.).
- Search Sources: We searched for articles containing the above keywords through Google Scholar and downloaded the articles that matched the requirements from the corresponding major databases.
- The articles we have studied include only high-level publications from international conferences and top journals (*e.g.*, EDM, KDE, SIGKDD, *etc*)based on the application of Deep Learning to educational scenarios.

Fig. 1 displays the distribution of collected papers categorized by publication year (grouping those before 2014, then annually) and across four key EDM application scenarios: knowledge tracing, student behavior detection, performance prediction, and personalized recommendation. The figure illustrates the proportion of studies within each scenario over time, highlighting that knowledge tracing and personalized recommendation consistently constitute the most substantial areas of research focus. In addition, it can be seen that EDM based on Deep Learning has emerged since 2015 [33].

Related Work. Ahmad et al. [34] examine the application of artificial intelligence in EDM, specifically focusing on the use of standard machine learning techniques for student assessment and prediction. Besides, Sun et al. [35] explore the application of Deep Learning in MOOCs for dropout prediction, focusing on the trends in using Deep Learning techniques, including feature processing and model design. Zhang et al. [36] also provide an overview of the educational data field, covering a range of classic Deep Learning models such as Stacked Auto-Encoder (SAE), Deep Belief Network (DBN), and various Deep Neural Networks (DNN). It further investigates distinct Deep Learning techniques tailored for different types of educational data, including large-scale, heterogeneous, real-time, and low-quality data.

Motivation and Contribution. Existing studies on the application of Deep Learning techniques in education are scattered and often lack a unified understanding of the techniques, models, and datasets. This motivates the need for a comprehensive survey to consolidate the current research, identify gaps, and provide a structured perspective on how Deep Learning algorithms can be better leveraged to enhance educational processes and outcomes.

This survey encompasses a diverse collection of state-of-the-art studies that categorize Deep Learning algorithms based on their application in educational settings. The articles are classified into three types: supervised learning, unsupervised learning, and reinforcement learning. Furthermore, the articles are further categorized based on the specific educational scenarios where Deep Learning techniques are employed. These scenarios encompass knowledge tracing, student behavior detection, performance prediction, and personalized recommendation. By adopting this systematic

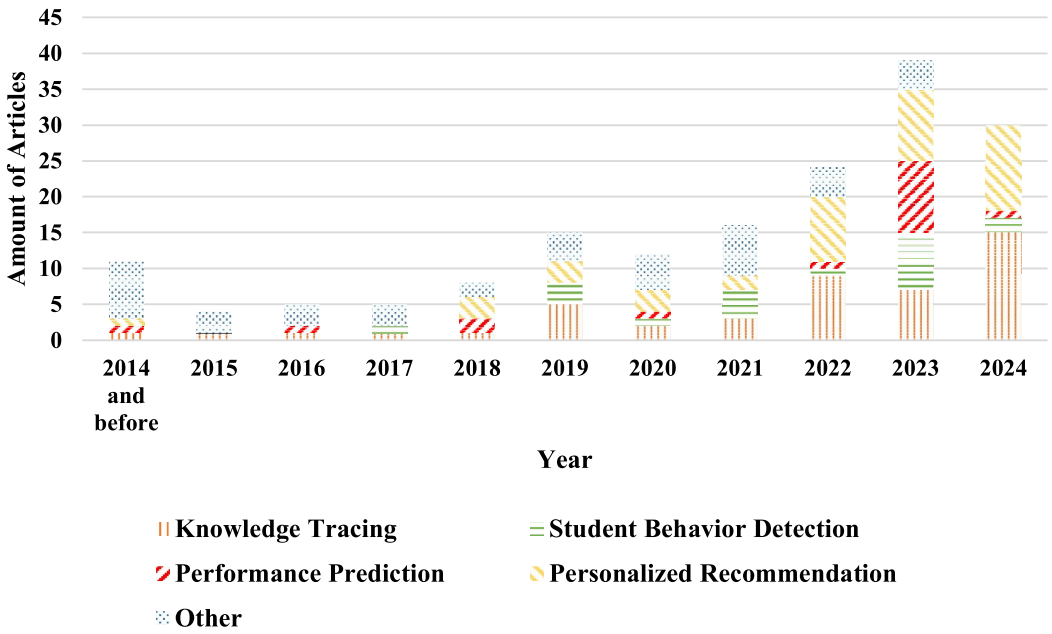


Fig. 1 Distribution of year-wise publications (until December 2024) according to different educational scenarios

approach, we provide a structured and in-depth analysis of the wide range of research conducted on the intersection of Deep Learning and education.

Different from other related work, this survey not only summarizes the latest Deep Learning algorithms but also includes a discussion of relevant datasets and data tools. By including a comprehensive analysis of datasets and data tools, the main objective of this survey is to offer the reader a thorough understanding of the considerations and challenges when implementing Deep Learning solutions in EDM. Furthermore, we put forward some promising areas for further research in Deep Learning-based EDM. The potential directions contain learning analysis and intervention, social network analysis and collaboration, Large Language Models for education, multimodal learning analytics, explainable Artificial Intelligence (AI) in EDM, scalability and real-time processing.

In summary, the purpose of this comprehensive survey is to provide a thorough and structured overview of the application of Deep Learning techniques in EDM. By categorizing and reviewing existing research, the survey aims to highlight key Deep Learning algorithms and their effectiveness in various educational environments. Additionally, the survey seeks to identify challenges, emerging trends, and future directions, offering valuable insights for researchers, educators, and developers looking to enhance educational outcomes through advanced data-driven approaches.

2 Educational Scenarios and Corresponding Algorithms

In this section, we summarize the Deep Learning techniques applied in knowledge tracing, student behavior detection, performance prediction, and personalized recommendation. Table 1 outlines common user behaviors and their key features across the four educational scenarios discussed. This allows researchers to choose Deep Learning techniques suited to these characteristics. For example, knowledge tracing focuses on student interaction sequences, which are typically temporal and sparse. Therefore, LSTMs can effectively model the sequential nature, while Transformers, via self-attention, can handle the sparsity. Student behavior detection focuses on user engagement and emotional expressions, data that is typically heterogeneous and noisy. Variants of complex networks can thus be used for handling heterogeneity, while GANs are effective for differentiating normal from anomalous behaviors. For the often encountered multimodal data, fusing different types of neural network helps to capture comprehensive information, and Transformers can process relevant contextual details. For performance prediction, core student behaviors involve performance changes, assignment submissions, and participation in interaction. Such data often has long-term dependencies and multiple influencing factors. RNN variants are effective for modeling these dependencies. As with student behavior detection, hybrid models are also useful here for handling complex multi-source or multimodal data. Personalized recommendation scenarios center on student choice paths: their content preferences and subsequent interactions (feedback, learning paths). This yields sparse sequential data. Thus, models must handle sequences effectively and address the cold-start problem; Transformers are a good fit. The scenario also frequently involves multiple objectives. Deep Reinforcement Learning can manage this via reward function design, making it another suitable choice.

To provide a comprehensive overview of the literature on Deep Learning for different application scenarios, we have included information on the algorithms and models used in each paper for four educational scenarios in

Table 1 Overview of user behaviors, key features, and available deep learning approaches for each educational scenario

Scenario	User behavior	Key feature	Main technology
Knowledge Tracing	Interact with Education Terms	Temporal and Sequential Knowledge State, Sparse Interaction Data	LSTM, Transformer
Student Behavior Detection	Platform Engagement	Patterns in Heterogeneous, Noisy Data	RNNs, LSTM, CNNs, GAN
	Sensing Affective State	Multimodal Information, Contextual Information	Multimodal Deep Learning, Transformer
Performance Prediction	Grades, Submissions, Participation	Long-term Dependencies, Multi-factor Impacts	RNNs, LSTM, Hybrid Models, GNN
Personalized Recommendation	Selective Behaviors, Interactions	Sparse Sequential Data, Balancing Exploration and Exploitation, Multi-objective Goals	Transformer, Deep Reinforcement Learning

Table 2 Overview of deep learning algorithms for different educational scenarios

Scenario	Model	Algorithm Classification	Method	Evaluation Metric	Dataset	Year
Knowledge Tracing	GFLDKT [37]	Supervised Learning	LSTM	AUC/ACC	ASSISTments15/17/JunyiAcademy	2023
	LFBKT [38]		LSTM	AUC/ACC	ASSISTment12	2022
	DFKT [39]		LSTM	AUC/ACC/MAP	ASSISTment12	2019
	DKVMN [40]		LSTM	AUC	ASSISTment09/15/Statics2011/Synthetic-5	2017
	PGN [41]		RNN	AUC/ACC/RMSE	ASSISTment15/17/Statics2011	2023
	DKT [33]		RNN	AUC	ASSISTment09/10	2015
	EERNN-EKT [42]		RNN	AUC/ACC/MAE/RMSE	iFLYTEK	2019
	KT-Deeper [43]		RNN	AUC	ASSISTment09/17/EdNet	2024
	GKT [44]		GNN	AUC	ASSISTment09/10/KDD Cup	2019
	DKT-STDRL [45]		CNN	AUC/ACC/RMSE/r^2	ASSISTment09/15/Statics2011	2022
	HHSKT [46]		GNN	AUC/ACC	ASSISTment09/17/Junyi15	2023
	TSKT [47]		GNN	AUC/ACC	ASSISTments09/15/Eanalyst	2024
	GIKT [48]		GCN	AUC	ASSISTment09/12/EdNet	2021
	FCIKT [49]		GCN/Attention	AUC	ASSISTments09/12/EdNet	2024
	MRTKT [50]		Attention	AUC/ACC	ASSISTment09/10/12/13	2023
	AKT [51]		Attention	AUC	ASSISTment09/10/12/13/KDD Cup2012	2020
	SAKT [52]		Attention	AUC	ASSISTment09/15/Statis2011	2019
	TCKT [53]		Attention	AUC/ACC/RMSE	ASSIST2012/ASSISTChall/EdNet	2024
	ARAIKT [54]		Attention	AUC/ACC	ASSISTments09/17	2024
	ELAKT [55]		Transformer	AUC/ACC/MAE/RMSE	ASSISTments09/10/15/17/EdNet	2024
	ENAS-KT [56]		Transformer	AUC/ACC/RMSE	EdNet/RAIEd2020	2024
	AdaptKT [57]	Unsupervised Learning	Autoencoder	AUC	zx.math/ax.physics	2022
	KTM [58]	Reinforcement Learning	AC	AUC	ASSISTment09/10	2019
	RL-KTNet [59]		AC	AUC/r^2	ASSISTment09/10/KKD2010	2020
	IEKT [60]		Policy Gradient	AUC/ACC	ASSISTment09/12/EdNet/Junyi	2021
	KADT [61]		DPG	AUC	ASSITment09/IMDB/MovieLens/CIFAR-100	2023
Student Behavior Detection	VB-DTW [62]	Supervised Learning	CNN	ACC/VSE	SCB-13	2023
	DCNN [63]		CNN	Recall/Precision/F1	FER2013	2023

Table 2 (continued)

Scenario	Model	Algorithm Classification	Method	Evaluation Metric	Dataset	Year
	ATGCN [64]		CNN	ACC	N/A	2023
	Faster R-CNN [65]		CNN	ACC/Precision/DR/FDR	TCE Classroom	2023
	CFIN [66]		CNN	AUC/F1	KDD Cup2015/XuetangX	2019
	EDLN [67]		CNN	AUC/Recall/Precision/F1	KDD Cup2015	2023
	ABDM [68]		CNN	ACC	N/A	2023
	LBDL [69]		LSTM	AUC/F1	MOOCCube	2023
	CDLSTM [70]		LSTM	AUC/Recall/Precision/F1	7WiseUp	2023
	ECLSTM [71]		LSTM	AUC/Recall/Precision/F1	KDD Cup2015	2024
	EVA-MLP [72]		MLP	Recall/Precision/F1	Student Journals	2023
	FTGAN [73]	Unsupervised Learning	GAN	AUC/ACC/Recall/Precision/F1	N/A	2021
Performance Prediction	DL-MLP [74]	Supervised Learning	MLP	ACC	Pennsylvania School Performance Profile	2023
	CRN [75]		CNN	FDR/Sens/FPR/MCC/F1/NPV/FNR	Kaggle	2023
	UPC-Net [76]		CNN	Recall/Precision/F1	Self-collected	2024
	SDPNN [77]		DNN	ACC	N/A	2023
	MLP-12Ns [78]		MLP	RMSE	Kaggle	2023
	TLBO-ML [79]		ANN	ACC/Recall/Precision/FM/F1/MCC	OULAD	2023
	SAPP [80]		LSTM	ACC/Recall/Precision/F-measure	OULAD	2023
	GRU-ANOVA [81]		RNN	ACC/Recall/Precision/F1	Self-collected	2024
	MSH-MD [82]		Attention	Recall/Precision/F1/MSE/MAE/RSME	Self-Collected	2023
	CGAN [83]	Unsupervised Learning	GAN	AUC	Self-collected	2022
	ADSLS [84]	Reinforcement Learning	Q-Learning	N/A	N/A	2013
	NCAT [85]		Q-Learning	AUC/ACC	ASSISTment09/12/15/KDD Cup2010	2022
Personalized Recommendation	BERT [86]	Supervised Learning	Attention	Recall/Precision/F1	MOOCCube	2023
	MCR-C-FGM [87]		DNN	Precision	edX Learning Data	2022
	SODNN [88]		DNN	Precision	UBOB	2023
	LSTM-CNN [89]		CNN	Recall/Precision/F1	Kaggle	2023
	CSEM-BCR [90]		CNN	Recall/Precision/MAP/AP	CourseTalk	2022
	ARGE [91]		RNN	AUC	LastFM/Movielens-100K/Yelp	2023
	KALUR [92]		GNN	Recall/NDCG	MovieLens-100K/Amazon-book/LFM-1B	2023
	PCGNN [93]		GNN	HR/MRR	Udemy1/XueTangX	2024
	GNN-SNA [94]		GNN	N/A	MOOCCube	2024

Table 2 (continued)

Scenario	Model	Algorithm Classification	Method	Evaluation Metric	Dataset	Year
	CR-LCRP [95]		HIN	AUC/NCDG/MRR/HR/AP	MOOCCube/MOOC review	2024
	TCRKDS [96]		LSTM	ACC/Recall/Precision/F-measure/FDR	Kaggle	2023
	FRS [97]		LSTM	AUC/Precision/RSME	ASSISTment	2018
	MRCRec [98]		GCN	HR/HR/NDCG/MRR	MOOCCube/XuetangX	2023
	ConceptGCN [99]		GCN/Transformer	Precision/Recall/F1	Inspec/SemEval2017	2024
	MRMLREC [100]		GCN/GAT/LSTM	HR/NDCG/MRR	Computer/MOOCCube	2024
	RecGAN [101]	Unsupervised Learning	GAN	NDCG/MRR/MAP	MPF	2018
	DBNLS [102]		DBN	Act/Ref/Sen/Int	StarC	2020
	RLALS [103]	Reinforcement Learning	AC	N/A	N/A	2018
	CSEAL [104]		AC	AUC/Recall/F1/MAP	Junyi	2019
	MEUR [105]		Q-Learning	IR/HR/NDCG/MRR	MOOCCube	2023
	QLearnRec [106]		Q-Learning	N/A	Self-Collected	2019
	RILS [107]		Q-Learning	N/A	Self-Collected	2014
	RPPR [108]		REINFORCE	HR/NDCG	MOOCCourse/MOOCCube	2023
	HELAR [109]		REINFORCE	HR/NDCG	MOOCCourse/MOOCCube	2022
	HRRL [110]		REINFORCE	HR/NDCG	MOOCCourse/MOOCCube	2022
	HRL-NAIS [111]		REINFORCE	HR/NDCG	MOOCCourse	2019
	KRRL [112]		AC	HR/NDCG	MOOCCourse/MOOCCube	2024
	UPGPR [113]		AC	NCDG/Recall/HR/Precision	COCO/XuetangX	2024

Table 2. This table also lists the evaluation metrics and datasets employed in each study.

2.1 Knowledge Tracing

Knowledge tracing [114] is an educational assessment technology that tracks students' learning process and predicts their mastery of knowledge points. In a knowledge tracing scenario, students' learning process is usually recorded, including learning time, answer status, homework completion, etc [115, 116].

According to previous studies proposed by Song et al. [22], the problem of knowledge tracing in intelligent education systems involves three main elements: the student S, the exercise E, and the corresponding knowledge concept C. The interactions X among these elements are the main activities in such systems. Specifically, given a student's historical exercise interactions $s \in S$, where each interaction $X_t \in X$ corresponds to an exercise $e \in E$ and denotes the correctness $a_t \in \{0, 1\}$ of the result obtained at step t, the knowledge tracing task aims to predict the next interaction X_{t+1} for a specific concept $c \in C$ [22].

A simple schema of knowledge tracing is shown in Fig. 2. The correct rate of a student in a certain exercise on a specific knowledge point can affect the model's judgment of the proficiency level of students on this knowledge point. For example, in Fig. 2, the student's correct rate on the derivative knowledge point is 100%, so his/her mastery level on that knowledge point is much higher than other knowledge points. However, traditional knowledge tracing methods often struggle with capturing the complex, sequential dependencies within student learning interactions and the nuanced relationships between knowledge concepts [117]. To address these limitations, the integration of Deep Learning techniques has emerged as a powerful approach.

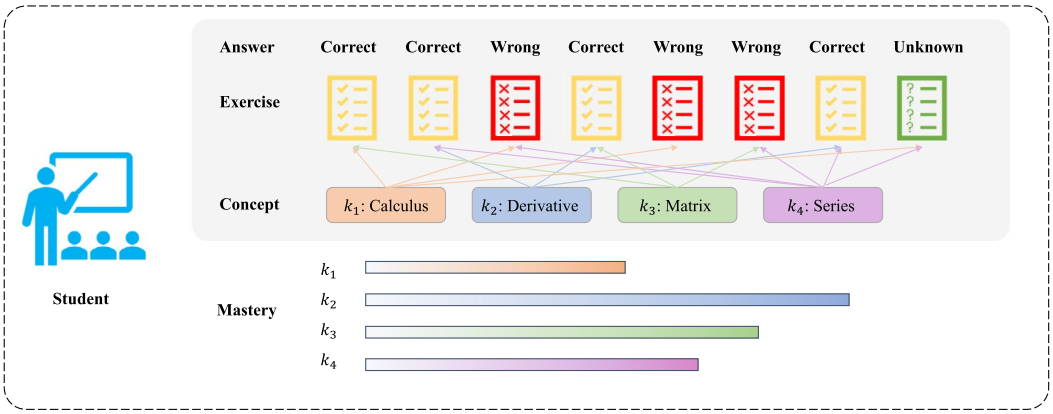


Fig. 2 A simple knowledge tracing schema. Different exercises contain multiple types of concepts with different colors. Whether the exercises are correct or not will affect knowledge tracing's judgment of the student's mastery of a certain knowledge point or exercise [118]

2.1.1 Supervised Learning

The application of supervised learning in the field of knowledge tracing is mainly focused on various kinds of neural networks, *e.g.*, LSTM [39], CNN [45], RNN [41], GNN [47] and Transformer [55, 56].

LSTM is primarily used to simulate the forgetting and learning processes of students, updating their knowledge states accordingly, thereby enabling effective knowledge tracing. A GFLDKT model based on LSTM was presented by Zhao et al. [37], in which a Gating-controlled Forgetting and Learning mechanism was employed to effectively update the knowledge state and facilitate accurate prediction of subsequent student responses. To illustrate how this Deep Learning architecture operates in a real-world EDM application, Zhao et al. [37] showcased GFLDKT's capabilities through a case study on the Junyi dataset [119]. By analyzing two students with differing initial abilities (one high-performing, s_1; one average, s_2) with identical early interaction sequences, the study demonstrated how the model discerns and understands distinct learning trajectories. For example, when faced with incorrect answers on the same concept ("Segments and Angles"), the learned gating parameters adjusted the knowledge state update differently for each student: s_1's mastery representation remained relatively stable, simulating resilience or learning from errors, while s_2's mastery level distinctly decreased. Furthermore, the Deep Learning model effectively captured personalized forgetting rates; the gating mechanism induced a significant decay in s_2's knowledge state for a concept ("Quadratics") not recently practiced, while having minimal impact on s_1's retention. The subsequent accurate prediction of both students' performance based on these dynamically computed internal states powerfully illustrates how GFLDKT leverages its gating components to capture nuanced, individual learning and forgetting patterns directly from interaction data.

In addition, Nagatani et al. [39] proposed a DFKT model considering the forgetting processes of the students. Specifically, DFKT applies LSTM and Neural Factorization Machine (NFM), the former is used to represent the knowledge state of the student as low dimensional dense vectors, and the latter combines student knowledge states and other related information including relevant forgetting data, to predict student performance. Compared to another LSTM-based knowledge tracing model tested on the same ASSISTment12 dataset [120], LFBKT [38] distinguishes itself in its approach to handling forgetting behavior. Unlike the DFKT model, which integrates forgetting data as part of its input, LFBKT opts for a more nuanced treatment by incorporating a dedicated Knowledge Forgetting Layer. This strategic design choice enables LFBKT to more effectively model the dynamics of knowledge decay over time. The impact of this approach is evident in the performance enhancement, with LFBKT's accuracy improving by 6.34% compared to the DFKT model on the ASSISTment12 dataset [120], demonstrating an advancement in knowledge tracing efficacy.

CNN also plays an essential role in knowledge tracing by being able to process the spatial sequence data. Lyu et al. [45] introduced a DKT-STDRL model that employs CNN to extract spatial features from students' learning sequences and LSTM to process temporal features.

Knowledge Graph (KG) is a crucial component of knowledge tracing, as it accounts for the interplay between a student's learning history and specific areas of expertise. By leveraging the power of the KG, models can more accurately capture learning trajectories and interactions between different points of knowledge. Yang et al. [48] put forth a Graph-based Interaction model for Knowledge Tracing (GIKT) constructed using Graph Convolutional Network (GCN). The proposed model addresses data sparsity and multi-skills

challenges by harnessing high-order question-skill correlations, thus improving model performance.

Moreover, a heterogeneous graph-based algorithm called HHKST was proposed by Ni et al. [46], which utilizes a GNN-based Base Feature Extractor (BFE) to extract interaction and knowledge structure features from the heterogeneous graph.

As a neural network architecture designed for handling sequential data, RNN also holds a significant position in the field of knowledge tracing. DKT [33] utilized considerable amounts of artificial neurons from RNN to construct a knowledge tracing model. The principle contribution of this article is the introduction of a novel method that encodes student interactions into RNN inputs and improves the AUC by 17% to 21% approximately compared to the Bayesian Knowledge Tracing (BKT) [121] model on three public datasets.

In knowledge tracing, the application of Attention mechanisms has been explored [53, 54]. Ghosh et al. [51] introduced a method called Context-Aware Attentive Knowledge Tracing (AKT), which combines a flexible DNN with heuristic cognitive and psychometric models. The proposed Attention mechanism is used to adjust predictions based on contextual information about learners dynamically. This integration of Attention allows the model to adapt its predictions according to the specific context of each learner.

Building on the concept of attention mechanisms, Cui et al. [50] proposed the MRTKT model as an innovative approach to knowledge tracing. Achieving an AUC of 82.23% on the ASSISTment09 [120] task, MRTKT slightly outperforms the AKT model, which scored 81.69%. Unlike AKT, which emphasizes context-aware dynamic adjustments using attention mechanisms, MRT-KT employs a multi-relational attention mechanism along with a relation encoding schema to enhance its predictive accuracy.

Additionally, Pandey and Karypis [52] proposed the Self-Attentive Knowledge Tracing model (SAKT), which leverages a self-attentive mechanism to identify and predict the mastery level of students for specific knowledge points. The experimental results demonstrate that SAKT outperforms traditional methods and RNN-based models, achieving significantly faster performance by an order of magnitude. Furthermore, the FCIKT model [49] utilizes GCN to obtain higher-order relationships from the constructed heterogeneous graph, and adopts the multi-head attention mechanism to learn the rich information from individual questions.

2.1.2 Unsupervised Learning

Although the preceding sections of our text have listed numerous knowledge tracing methods based on supervised learning, strictly classifying scenes into specific algorithm categories (Supervised, Unsupervised, reinforcement learning) is often challenging. While traditional methods like BKT [121] are seen as unsupervised learning, they don't incorporate Deep Learning techniques. Hence, applying unsupervised Deep Learning in knowledge tracing is a relatively new and developing area. However, in recent years, with the rapid development of Deep Learning, some researchers have started to explore the use of unsupervised learning in knowledge tracing. For instance, Cheng et al. [57] proposed a Autoencoder- based DKT, which combines knowledge tracing and transfer learning. The Autoencoder here is used to convert question text to high-level semantic embedding:

$$\begin{aligned} &\text{Encoder}: q = \pi_e(x), \\ &\text{Decoder}: \hat{x} = \pi_d(q), \end{aligned} \tag{1}$$

where x refers to the question texts needed to be encoded, π_e is the Encoder function, $\hat{x}$ stands for the reconstruction from q through Decoder function π_d. Here, the model applies Bi-LSTM and Attention mechanism as Encoder and Decoder functions, to better capture the knowledge state of the student and predicts the next answer of the learner by Softmax.

2.1.3 Reinforcement Learning

Reinforcement learning is able to learn the knowledge status and level of students by rewarding or punishing them through decisions and actions they make. The learning process in knowledge tracing can be considered as a sequential decision problem, students need to make different decisions according to different knowledge points, which might affect the future learning process.

Ding et al. [59] considered that some supervised learning, such as LSTM or GRU, are heavily influenced by NLP but not specially designed for knowledge tracing. Thus, the authors design an RL-KTNet algorithm that applies reinforcement learning to automatically generate RNN cells used in knowledge tracing. It outperforms other models employing LSTM cells in terms of AUC.

In addition, Long et al. [60] proposed a model called Individual Estimation Knowledge Tracing (IEKT), which incorporates reinforcement learning for auxiliary model training. Specifically, Long et al. employed Policy Gradient and ϵ -greedy to update the model parameters. Additionally, reinforcement learning is utilized to estimate student knowledge states and sensitivity toward knowledge acquisition.

As shown in Fig. 1, the application of reinforcement learning in knowledge tracing has been increasing in recent years. Girard et al. [122] explored how reinforcement learning can optimize human learning, aiming to learn instructional policies from student data to understand how students learn and which exercises significantly impact learning. Another study [123] focused on the sequencing of questions

or lectures, providing an end-to-end pipeline to derive and evaluate robust student representations and content sequencing strategies from student interaction data. Besides, Zhang et al. [124] proposed a framework that combines reinforcement learning and KG to model the learning path recommendation problem as a sequential decision-making process, and students gradually construct the knowledge state through interaction with learning resources.

In summary, Deep Learning offers diverse knowledge tracing methods, each with specific strengths suitable for different educational contexts. Sequential models like LSTMs effectively track learning progression and forgetting, as demonstrated by LFBKT [38], which introduced a dedicated forgetting layer improving upon earlier LSTM approaches like DFKT [39]. However, these models, including foundational RNN-based methods like DKT [33], can sometimes struggle to capture complex, long-range dependencies within learning sequences. Attention mechanisms and Transformers address this limitation; for instance, AKT [51] incorporates context-aware attention, while SAKT [52] utilizes self-attention for efficiency gains over traditional RNNs. Graph Neural Networks (GNNs) provide a distinct advantage when modeling the explicit structure of knowledge concepts and their relationships, as seen in GIKT [48] using GCNs or HHKST [46] employing heterogeneous graphs, proving particularly useful for handling data sparsity but dependent on accurate graph construction. Other techniques also contribute; for example, unsupervised methods like Autoencoders in AdaptKT [57] aid in learning feature representations, while reinforcement learning, used in models like IEKT [60], uniquely focuses on optimizing learning policies rather than just prediction. Therefore, selecting an appropriate KT model requires balancing the benefits of temporal tracking (LSTMs/RNNs), structural modeling (GNNs), contextual understanding (Transformers), or policy optimization (RL) against the specific pedagogical goals, data characteristics, and computational constraints of the application.

2.2 Student Behavior Detection

In the current era, both traditional and online educational systems have been generating massive amounts of data. The extraction of useful knowledge and underlying patterns from this voluminous data can enable decision-makers to enhance teaching and learning by identifying student behaviors [25]. This educational data can be considered as an invaluable source of information that can facilitate data-driven educational research and innovation. The objective is to identify student behaviors such as low motivation, low engagement, cheating, dropout and procrastination [125]. However, effectively leveraging the potential of this data presents significant challenges. Traditional analytical methods often struggle with the high dimensionality, heterogeneity, and sheer volume of educational data. Furthermore, capturing the complex, non-linear relationships and temporal dynamics inherent in student learning processes can be particularly difficult, limiting the accuracy and depth of insights derived from simpler models.

2.2.1 Supervised Learning

The applications of supervised learning in this field are relatively diverse, variety of neural networks are applied, such as CNN and LSTM.

CNNs are frequently used in applications that require spatial feature extraction from image or video data due to their effectiveness in such tasks. Wang et al. [62] implemented CNNs to analyze data obtained from motion sensors, leading to accurate recognition of 14 common classroom behaviors.

Leveraging CNN's superior image processing, Hdioud et al. [63] developed a DCNN model for recognizing students' faces with masks during the epidemic, showcasing adaptability under challenging conditions. Concurrently, Qiu et al. [64] proposed an ATGCN model, also based on CNN, to identify improper behaviors like napping in classrooms. This highlights the diverse applications of CNN in educational settings, where one model focuses on facial recognition under constraints and the other on behavior detection. The ATGCN leverages both CNN and GCN as its fundamental components and employs an attention mechanism to boost its performance. A concise representation of the ATGCN structure is depicted in Fig. 3.

In education scenarios, student behaviors extend beyond simple actions such as napping and inattentiveness. An important part of this field is the detection of dropout tendencies, which is integral to understanding the full scope of student behaviors. Feng et al. [66] proposed a Context-aware Feature Interaction Network (CFIN) model for predicting students' dropout behaviors in MOOCs. The CFIN model incorporates a context smoothing technique to enhance the feature values across different contexts and employs an attention mechanism to integrate user and course information within the modeling framework.

Furthermore, Kumar et al. [67] developed an EDLN model that combines CNN and an attention mechanism for student dropout detection in online courses. The EDLN model employs ResNet-50 to extract local high-dimensional features and uses Faster R-CNN to analyze hidden long-term memory features in time series data. This model demonstrates notable performance, achieving an accuracy of 97.5% with a 5×7 time series matrix as input.

Other deep neural networks can also be employed for dropout prediction. The LBDL model, as proposed by Liu et al. [69], integrates Bi-LSTM and a multi-head attention mechanism to analyze time series information extracted from

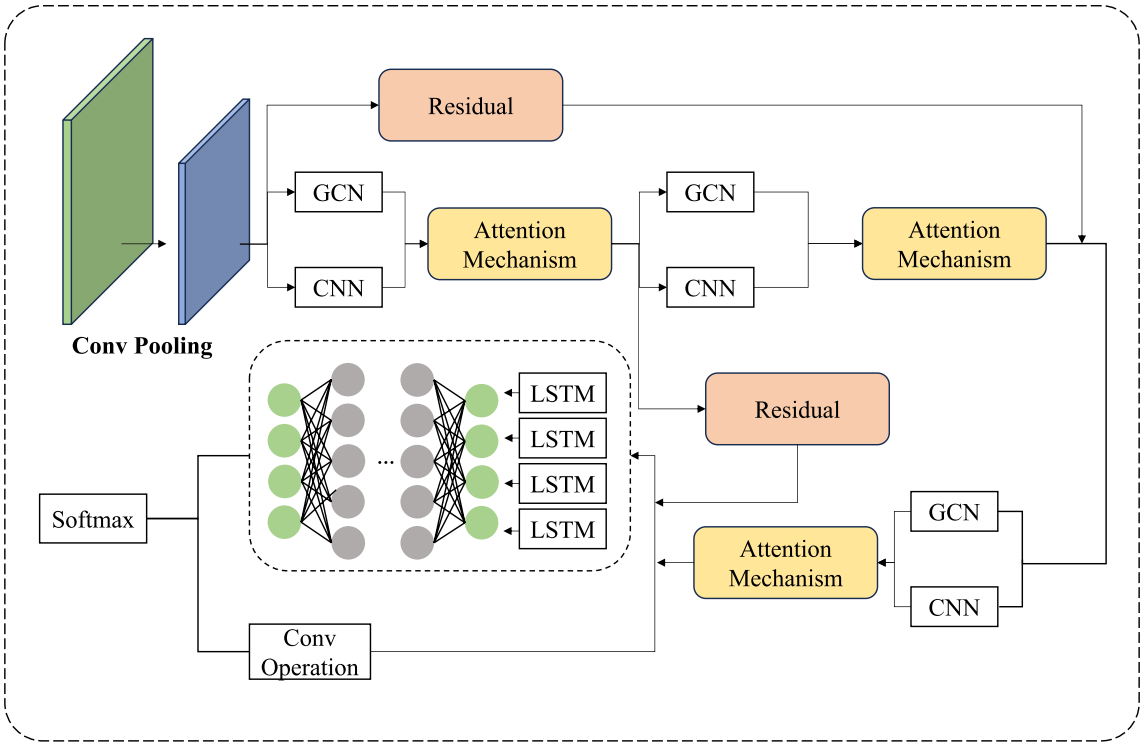


Fig. 3 A simple structure of ATGCN model [64] for student behavior detection

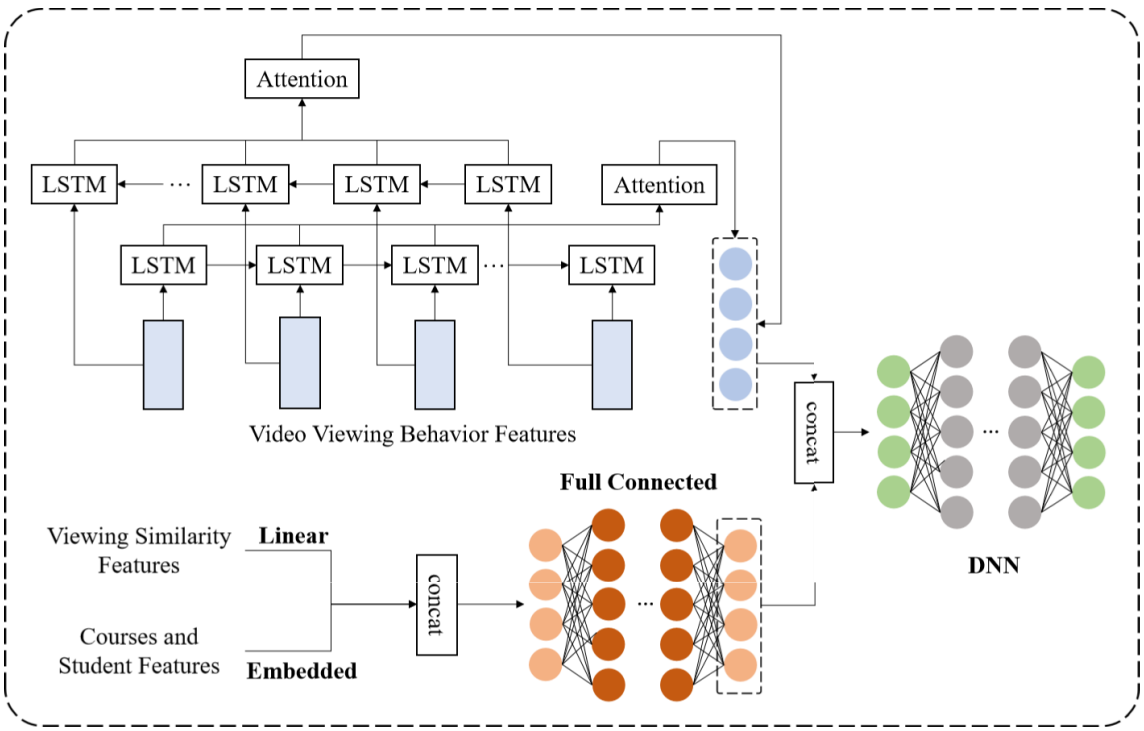


Fig. 4 A simple structure of LBDL model [69] for dropout prediction

video-based study behaviors. This model demonstrates performance with AUC and F1 scores of 82.39% and 74.89%, respectively, on the MOOCCube dataset [126]. A simple structure of LBDL is shown in Fig. 4. The model involves two stages shown in Fig. 4. First, the left side uses Deep Learning models to learn hidden feature vectors from the input data. Second, the bottom right utilizes these learned hidden features, plus original information, to train the final LightGBM predictor. Subsequently, the authors demonstrate the exceptional performance of this model [69].

With the advancement of online education, the detection of cheating has become a critical area of focus for EDM researchers. Alsabhan et al. [70] developed the CDLSTM model, which is based on the LSTM. This model incorporates a dropout layer, a dense layer, and the Adam optimizer. Owing to the LSTM's ability to effectively process sequential data, the CDLSTM model achieved a notable accuracy of 90% in identifying cheating behaviors, using students' online test activity logs as input.

2.2.2 Unsupervised Learning

In this section, we delve into the application of unsupervised learning within EDM, particularly in detecting student behaviors. This approach, a facet of Deep Learning, processes educational data to autonomously identify patterns in student engagement, providing insights without relying on predefined labels or categories.

Stenton et al. [73] introduced an FTGAN which means Fine-Tuning GAN to predict the attrition rate of the student. GAN constructs the basis of this approach, where the interplay between a discriminator model and a generator model is key. This dynamic is captured by the GAN's value function:

$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{data}(\mathbf{x})}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))], \tag{2}$$

where G is the generator, D is the discriminator, $\mathbf{x}$ is real data, $\mathbf{z}$ is generator input noise, and $\mathbb{E}$ is the expected value. The authors demonstrate that the more epochs the GAN classifier model is trained, the more its accuracy shows a certain level of increase.

2.2.3 Reinforcement Learning

While exploring the application of various algorithms in detecting student behaviors, it is crucial to consider the complexity and multifaceted nature of these behaviors. Reinforcement learning, though powerful, faces challenges in this domain.

- The complexity of student behavior, influenced by diverse factors like social background and personal circumstances, may not align well with the straightforward state-action framework typically used in reinforcement learning.
- Defining appropriate rewards and punishments in the context of student behaviors and their long-term outcomes, such as dropout rates, presents a significant challenge. The dichotomy between short-term and long-term consequences complicates the application of reinforcement learning.

Based on the outlined reasons, our study does not include literature on the application of reinforcement learning for student behavior detection. However, we acknowledge the potential value and relevance of attempts to explore this method in such contexts.

In summary, Deep Learning effectively detects student behaviors in complex data where traditional methods falter. CNNs excel with spatial/visual inputs (*e.g.*, classroom actions [62]), but may be less optimal for analyzing purely sequential, non-visual patterns. LSTMs effectively process sequential data like activity logs [70] or video analysis [69], though standard architectures can struggle with very long-range dependencies if not enhanced (*e.g.*, with attention [66]). Unsupervised methods like GANs [73] show potential but are less established, suggesting current limitations in robustness or general applicability for diverse detection tasks. Reinforcement learning is seldom used for behavior detection due to fundamental challenges in defining appropriate states and reward functions for complex human behaviors. Therefore, selecting a model requires weighing its advantages (CNN-spatial, LSTM-sequential) against potential limitations and the specific nature of the behavior and data being analyzed.

2.3 Performance Prediction

Performance prediction [26] refers to the use of various data and analytic techniques to predict student performance in certain tasks or areas, such as test scores, academic performance, course completion rates, etc. The difference between it and the knowledge tracking task is that performance prediction focuses on predicting students' overall future tasks or test performance based on historical learning data, while knowledge tracing focuses on students' understanding and mastery of specific knowledge concepts during the learning process.

By predicting student performance, teachers and educational institutions can better understand students' learning and needs to provide more effective support and guidance. Predictions can also help students understand their performance and potential difficulties and take steps to improve learning outcomes.

Furthermore, cognitive diagnosis [127] is a related concept wherein the aim is to identify students' level of mastery in specific knowledge domains by analyzing their performance on exercise records. This analysis facilitates providing tailored guidance for their subsequent studies [128, 129]. In contrast, performance prediction focuses more on predicting students' overall scores on tests. The former analyzes knowledge dimensions, while the latter emphasizes general ability. Cognitive diagnosis outputs concept proficiency levels, whereas performance prediction directly predicts total scores. However, achieving accurate performance prediction and cognitive diagnosis faces challenges such as data sparsity, where interaction data for students across all concepts or tasks is often incomplete. Capturing the dynamic changes in student abilities and knowledge mastery over time, and precisely modeling the complex interplay between different knowledge components and overall performance, also remain significant hurdles for traditional methods.

2.3.1 Supervised Learning

Supervised learning algorithms can provide teachers and students with useful information to help them better understand student performance and needs. In addition to classic algorithms like SVM, Deep Learning models such as MLP, CNN, RNN, and LSTM have been applied to predict student performance in various educational contexts.

Nayani et al. [75] introduced a hybrid model called CRN, which is a combination of CNN and RNN to predict the students' grades and improve the performance by tuning hyperparameters through Galactic Rider Swarm Optimization (GRSO) algorithms.

In addition, Neha et al. [77] presented a SDPNN model that applied a linear classifier-based DNN to predict student academic performance. There are two hidden layers with 300 neurons defined. The activation function is ReLU and Softmax.

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}}, \quad i = 1, 2, ..., K, \tag{3}$$

where z_i represents the i_{th} element of the input vector, and K is the number of categories.

Kukkar et al. [80] proposed a SAPP system that utilize four layers of stacked LSTM, Random Forest and Gradient Boosting. Here, LSTM is used to extract features while Random Forest and Gradient Boosting are applied to predict. The proposed system has an accuracy rate of 96% and performs well to some extent.

Furthermore, Chen et al. [82] introduced the MSH-MD model, which also utilizes LSTM. The core of MSH-MD is its self-attention mechanism, integrating LSTM's capacity for processing time series with the efficient feature extraction of the self-attention mechanism. Compared to the SAPP model [80], the MSH-MD focuses more on predicting student performance through pattern differences, whereas the SAPP model emphasizes feature extraction more prominently.

2.3.2 Unsupervised Learning

Although the grade prediction usually relies on labeled datasets for supervised learning, some unsupervised learning algorithms, such as GAN, can also be used for grade prediction tasks. GAN can perform grade prediction by taking a student's historical grades as input and using generators and discriminators to generate predicted values for future grades. The generator can use the student's prior grades and other relevant factors to generate predictions of future grades, and the discriminator is used to determine whether the generated predictions are similar to the true grades.

Sarwat et al. [83] proposed a model combined with Conditional GAN (CGAN) and Deep-Layer-based SVM to predict students' grades according to school or home tutoring. CGAN was used to generate performance score data to address the issue of small dataset size, and the model using a combination of CGAN and SVM was experimentally shown to have a positive effect on the prediction results.

2.3.3 Reinforcement Learning

Generally speaking, while reinforcement learning may not be traditionally used for student performance prediction, it can be employed to optimize a student's learning path by setting reward strategies based on student behavior. In this context, reinforcement learning would also involve assessing the current performance of students and considering potential improvements.

Dorça et al. [84] proposed an ADSLS model to automatically detect and precisely adjust students' learning styles. An important part of this is that the model predicts and assesses student performance on a point and rewards performance while updating learning strategies. The proposed method's efficacy and efficiency have been demonstrated by the results.

Zhuang et al. [85] introduced an NCAT model that utilizes reinforcement learning to enhance the effectiveness of e-testing systems. The NCAT model employs deep reinforcement learning to dynamically optimize test terms based on given conditions. With the proposed algorithms, the e-testing system can provide more comprehensive and accurate performance predictions.

Overall, Deep Learning offers powerful methods for this scenario, but they come with their limitations. For example,

supervised LSTM models excel at processing sequential performance data, used for feature extraction (SAPP [80]) or combined with attention for deeper pattern analysis (MSH-MD [82]); however, they can be computationally expensive and require careful tuning to effectively capture all relevant long-term dependencies. Hybrid models combining CNNs and RNNs (CRN [75]) capture diverse features but add optimization complexity. Simpler DNN models (SDPNN [77]) risk oversimplifying the complexity of learning progressions. Furthermore, overfitting is a common risk when dealing with limited or sparse data, and the 'black-box' nature of complex models limits the interpretability of predictions. Unsupervised methods, like CGANs [83], primarily address data scarcity through augmentation-a useful supporting role, but they do not directly predict scores themselves, and the quality of generated data is critical. Reinforcement learning is typically not used for direct prediction but applied to optimize related processes, such as adaptive testing (NCAT [85]) or learning style adaptation (ADSLS [84]); these processes involve performance assessment but face challenges in state-space and reward function definition. Therefore, applying Deep Learning in student performance prediction requires selecting models (like LSTMs for sequences or hybrids for feature integration) while actively mitigating risks such as overfitting, computational cost, and lack of interpretability, ensuring the chosen approach aligns with data availability and specific predictive goals.

2.4 Personalized Recommendation

In the current era of explosive growth in online information, recommendation systems [28, 130] undoubtedly offer an effective means of addressing this issue and providing necessary assistance to individual users. Even outside the educational context, recommendation systems remain one of the most widely studied technological approaches. Zhang et al. [131] gave a detailed definition of a recommendation system: Assuming the presence of M users and N items, we denote the interaction matrix and predicted interaction matrix as R and $\hat{R}$, respectively. The user preference for item i is represented as r_{ui}, while the predicted score is denoted by $\hat{r}_{ui}$. There will also be two partially observed vectors, one represents a specific user u, *i.e.*, $r^{(u)} = \{r^{u1}, ..., r^{uN}\}$. The other one represents a specific item i, *i.e.*, $r^{(i)} = \{r^{1i}, ..., r^{Mi}\}$.

In the education scenarios, the item is replaced with educational resources such as courses. The primary objective of a course recommendation system is to suggest the most suitable course to a user at time $t + 1$, taking into account their past learning activities and learner profiles prior to time t. The primary challenge faced by such recommendation systems is to provide personalized recommendations by precisely depicting and conceptualizing user inclinations through analysis of user data [132]. Fig. 5 illustrates a typical personalized recommendation loop in EDM. User and Course data are modeled, often using Deep Learning, to inform a central recommendation engine. This engine generates suggestions for the user, and user feedback subsequently updates the models for continuous improvement.

2.4.1 Supervised Learning

The application of supervised learning in course recommendation [87] focuses on a wide variety of neural network models due to the greater flexibility and expressiveness of neural networks, which can better handle user behavior sequences and nonlinear features. At the same time, neural

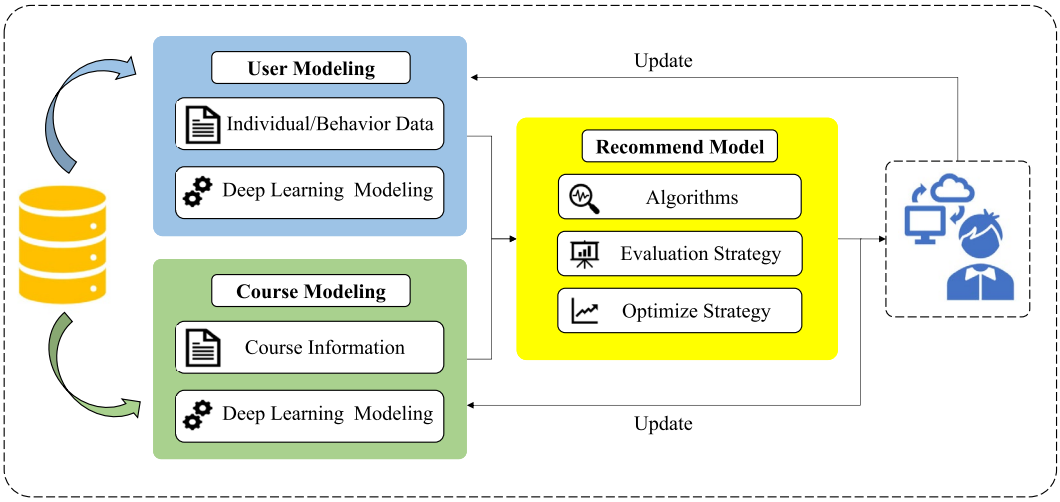


Fig. 5 A general process framework for personalized recommendation in EDM. The recommendation system uses user data and course data for modeling. After recommending to users, the associated data between users and courses will be updated to improve accuracy

networks can automatically extract features from the data to more accurately predict users' interests in the recommendation process.

Some optimized neural network models play a role in the field of personalized recommendation. SODNN, a novel model consisting of synchronous sequences, heterogeneous features and DNN has been introduced by Safarov et al. [88]. At the same time, to solve the cold-start problem, *i.e.*, Large errors caused by missing user data during the initialization of the recommendation system, the authors tried to concatenate additional features to overcome it.

Furthermore, classical neural networks can also be employed in recommendation systems. In contrast to the conventional positive sequence modeling approach, Gao et al. [90] proposed a novel CSEM-BCR model that adopts negative sequence modeling. Specifically, this model constructs the course-learning sequence as a negative sequence pattern, in which the negative term refers to the principle that students should not select or operate courses inappropriately. The negative sequence pattern is then processed using CNN for feature learning, which generates a list of recommended courses for each user. The suggested approach presents a fresh perspective on personalized recommendation and offers a potential solution to the issue of recommending courses to learners with different needs and preferences.

KG is an important part of personalized recommendations. It can effectively address the sparsity issue in recommender systems. An ARGE model based on multiple paths RNN encoder is proposed by Zhao et al. [91]. The model solves the problem that traditional RNNs do not consider the association between paths for encoding, and the AUC and Precision in the experimental outcomes demonstrate the model's capacity to effectively address the issue of sparse interaction between users and items.

GNN, as a neural network model specifically for processing graph structures, has also been applied by many scholars to deal with recommender systems [93, 95]. Ma et al. [92] presented a KAUR model that applies GNN to learn node representations for each node in the collaborative KG. The model treats the node information and its neighboring node information that has been propagated as positive contrastive pairs and then leverages contrastive learning to improve the quality of the node representations.

Zhou et al. [97] proposed a Full-path Recommendation System (FRS) based on LSTM and a clustering algorithm. The clustering algorithm is utilized to categorize learners based on their similar learning features, which in turn helps classify the learning paths according to the previous results. Consequently, this approach effectively addresses the cold start problem. LSTM is employed for predicting learning performance, and if the result is unsatisfactory, the system will select the most relevant learning path for users based on their individual learning features.

Some researchers leveraged the hybrid methods [100] to simultaneously address multiple issues or challenges in course recommender systems. For example, Alatrash et al. [99] proposed a ConceptGCN framework to enhance the e-learning experience. It combines propagation-based GCN with KG to effectively represent items (*e.g.*, courses, videos, and knowledge concepts), and then adopts Transformer to better capture the importance of relations between different knowledge concepts. Finally, it uses these enhanced representations to construct learner models, facilitating more personalized recommendations of associated knowledge concepts.

2.4.2 Unsupervised Learning

In recommender systems, user preferences and behaviors are often incomplete and inaccurate, and tagging data is very difficult and expensive to collect. On this basis, unsupervised learning is a great means to achieve personalized recommendations. Unsupervised learning can extract potential interests and preferences from users' historical behaviors and infer the similarity and interest relevance of users through clustering and feature learning to achieve personalized recommendations.

Bharadhwaj et al. [101] introduced a hybrid RecGAN model based on GAN and RNN. The generator and discriminator are both constructed with GRU-based RNN. The generator is then allowed to play a Mini-Max game with the discriminator, *i.e.*, there exists a true distribution $D_{real|t}$ in time index t, and a probability distribution generated by the generator $D_{gen|t}$. The goal of this minimal-maximization game is to minimize the generation error of the generator, while maximizing the discriminator's ability to distinguish false ratings from true ratings.

Unsupervised learning can also be applied to classify different types of learning styles to recommend the most suitable educational resources to distinct groups. A DBN-based DBNLS model proposed by Zhang et al. [102] is used to detect and classify learning styles. The core component of DBNLS is the multilayer RBM and Back Propagation network layer, where Back Propagation is used to fit the DBNLS model and to fine-tune and train the DBN. The energy function of RBM is:

$$E_\theta = -\sum_{i=1}^{n_v} b_i v_i - \sum_{j=1}^{n_h} a_j h_j - \sum_{i=1}^{n_v}\sum_{j=1}^{n_h} w_{ij} v_j h_j, \tag{4}$$

where v is the state vector for the visible layer, h is the state vector for the hidden layer, $a_j \in a$ is the bias of the i neurons in the visible layer, $b_i \in b$ represents the bias of the j neurons in the hidden layer, and $w_{ij} \in W$ is the element of the weight matrix between the j-th hidden layer and the i-th visible layer.

2.4.3 Reinforcement Learning

The application of reinforcement learning in personalized recommendation has high research value and development prospects. Reinforcement learning algorithms can learn the optimal recommendation strategy based on the learner's feedback to improve the recommendation performance.

A cognitive structure enhanced framework for adaptive learning named CSEAL was proposed by Liu et al. [104], to achieve personalized learning path recommendations. This framework views the learning path as a Markov Decision Process (MDP) and applies actor-critic to identify appropriate learning projects for individual learners. CSEAL comprehensively considers the learner's knowledge level and the knowledge structure of the learning project. Experimental results demonstrate that CSEAL can enhance learners' efficiency compared with current adaptive learning methods.

Liang et al. [105] presented a MEUR model based on Graph Convolution Network and reinforcement learning. The author considered the learning process as a MDP. Applying a user-centric reasoning method and maximizing cumulative reward G by actor-critic algorithm, G is defined as below:

$$G(\theta) = \frac{1}{m} \sum_{u=1}^{m} \sum_{a,t=0} \pi(a|s_t, A(s_t)) \gamma^t R(s_{t+1}), \tag{5}$$

where θ is the parameter of the Actor network, m is the number of samples, u is the sample index, a and t are the time step and action indexes, $\pi(a|s_t, A(s_t))$ is the probability of selecting action a in state s_t, $R(s_{t+1})$ is the immediate reward obtained in state s_{t+1}, and γ is the tuning parameter.

It is important to propose explainable recommendation methods in MOOC for interpreting the recommendation results. To this end, Lin et al. [112] developed a KRRL model, combined KG and self-supervised reinforcement learning, for explainable MOOC recommendation. Specifically, a multi-granularity representation learning method enriches the perceptual information of semantic interactions in the KG, and a self-supervised reinforcement learning method guides the path reasoning over the KG, and finally recommends appropriate courses to the target learner. The experimental results show that KRRL performs better than some comparison methods, in terms of the recommendation accuracy and explainability. Similarly, the UPGPR model [113] also utilizes the KG and RL to implement explainable recommendation in MOOCs. UPGPR, which stands for Unrestricted Policy-Guided Path Reasoning, is an algorithm that generates recommendations by identifying optimal paths within a KG that connect learners to potential courses. Unlike its predecessor (PGPR), UPGPR allows for more flexible path patterns, enhancing its generalizability. This adaptability enables UPGPR to provide explanations for recommendations by showing the connections between the learner's history and the suggested courses. In evaluations, UPGPR has demonstrated competitive accuracy compared to state-of-the-art MOOC recommendation models, while also offering the benefit of interpretable, path-based explanations.

Besides, reinforcement learning algorithms can be used to model learner preferences. For example, Zhang et al. [111] pioneered the use of Hierarchical Reinforcement Learning (HRL) to refine learner profiles. Their method addresses the issue where diverse course enrollments can dilute relevant courses on recommendations. They employ an HRL-driven profile reviser to selectively remove less relevant courses, enhancing the accuracy of a basic recommendation model. Experiments on the XuetangX [66] platform validated that this approach outperforms existing methods. Furthermore, the HRRL algorithm [110] integrates Recurrent Reinforcement Learning (RRL) with HRL as a profile reviser, which iteratively revises learner profiles to assist the MOOC recommendation model. Lin et al. [108] also proposed an RPPR model, in which a multi-scale reinforcement learning method constructs multi-dimensional learner profiles, thereby dealing with the adverse effects of insufficient semantic information on learner modeling.

In summary, Deep Learning provides diverse methods for personalized educational recommendation, addressing challenges such as data sparsity and preference modeling, although each approach involves trade-offs. Supervised techniques, utilizing LSTMs/RNNs [91] for sequences or GNNs [92] for structural data (often relying on KGs), offer flexibility but can be complex and data-dependent. Hybrid models [99] enhance performance at the cost of increased complexity. Unsupervised methods like GANs [101] are useful with unlabeled data but may face training stability issues, while others provide indirect personalization [102]. Reinforcement learning excels at optimizing recommendation sequences [104] and can offer explainability [112], but entails significant design complexity regarding states, actions, and rewards. Therefore, selecting an approach requires balancing modeling strengths against limitations concerning complexity, data needs, stability, and explainability requirements.

3 Datasets and Processing Tools

3.1 Datasets

To provide a more comprehensive overview of commonly used public datasets in educational settings, we have curated

Table 3 Dataset information

Dataset name	Description	Application scenario	Literature applied
ASSITments [120][1]	A knowledge tracing dataset based on knowledge decomposition theory in education, including problem-solving records. Aimed for developing and evaluating intelligent education systems	Used for researching and evaluating intelligent education systems based on knowledge decomposition theory, such as knowledge tracing and personalized recommendation, using machine learning and Deep Learning techniques	[33] [37] [46] [133] [41] [134] [38] [135] [45] [50] [85] [48] [127] [60] [51] [52] [136]
Junyi Academy Online Learning Activity Dataset [119][2]	Junyi dataset is an educational dataset that contains a large amount of online learning behavior data of primary and secondary school students, including learning and answering records, learning behavior features, and other information	The dataset is designed to support data mining task, *e.g.*, knowledge tracing	[37] [46] [133] [41] [104] [97] [60]
KDD Cup 2010 [137][3]	This dataset contains logs of student interactions with computer-aided-tutoring systems, including problem-solving transactions. Key terms include problem, step and knowledge component	It can be applied to the development and evaluation of intelligent educational systems, personalized recommendation, knowledge tracing and Deep Learning-based educational technologies	[134] [138] [139] [140] [141]
MOOCCube [126][4]	MOOCCube is an open data warehouse for NLP, KG, and data mining researchers in large-scale online education. It includes 706 real online courses, 38,181 instructional videos, 114,563 concepts, and 199,199 MOOC users' hundreds of thousands of course selection and video-watching records	The data set can be used to study learner behavior patterns, which can be utilized for personalized recommendation	[142] [98] [143] [105] [144] [109] [110] [69]
xAPI-Educational Mining Dataset [145, 146][5]	This dataset contains 480 student records and 16 features, including demographic attributes (gender, nationality), academic background information (educational stage, grade level, section), and behavioral indicators (raising hand in class, accessing resources, parental survey responses, school satisfaction)	xAPI-educational mining dataset can be used to develop and evaluate learning analytics models, to implement specific tasks such as personalized recommendation, performance prediction and student behavior detection in the education domain	[101] [147] [148]
Open University Learning Analytics Dataset [149][6]	The Open University Learning Analytics Dataset is a comprehensive educational dataset that includes over 300,000 students' demographic information, course information, interaction data, and achievement data from seven courses, and has been used in various educational research studies	Predicting student success and dropout rates in higher education, as well as understanding learning behavior patterns in online courses. So as to develop personalized recommendation and performance prediction system	[80] [96] [79]
Canvas Network Dataset [150][7]	Online courses data collected from Canvas Network platform, including course metadata, enrollment, and interaction logs	Predicting and understanding student performance in online courses, and designing personalized recommendation and student behavior detection model	[151] [152] [153] [154] [155]

Table 3 (continued)

Dataset name	Description	Application scenario	Literature applied
Learn Moodle [156][8]	The Learn Moodle dataset is a collection of student activity logs, discussions, quizzes, and more from various Moodle courses. It is used for researching student behavior patterns, learning outcomes, course design, and Moodle platform functionality in online education	This dataset is mainly used to study student behavior patterns, learning outcomes, course design, and Moodle platform functionalities in online education, aiming to improve teaching quality and efficiency in online education	[157]
XuetangX [158][9]	XuetangX is a Chinese dataset, which covers courses and learning behavior data of Chinese users on the online platform of XuetangX. This includes student registration information, course access records, video viewing behavior, homework submission, discussion participation, etc	The data set can be used in the research and application of learner behavior analysis, knowledge tracing and personalized recommendation	[111] [98] [105] [144] [109] [110] [159] [66]
EdNet [160][10]	EdNet is a large-scale educational dataset collected from Santa, an AI tutoring service with over 780 K users. It contains multivariate data on student interactions across platforms, including materials consumed, responses given, and time spent on learning activities. The key properties are its large scale, detailed student behavior data, and collection from a deployed system with numerous real users	EdNet enables personalized recommendation, knowledge tracing and educational optimization through its large-scale, detailed data on student behaviors and interactions	[50] [48] [60] [55] [53] [56]

[1] (https://sites.google.com/view/assistmentsdatamining/home) [2] (https://www.kaggle.com/datasets/junyiacademy/learning-activity-public-dataset-by-junyi-academy) [3] (https://pslcdatashop.web.cmu.edu/KDDCup) [4] (http://moocdata.cn/data/MOOCCube) [5] (https://www.kaggle.com/datasets/aljarah/xAPI-Edu-Data) [6] (https://analyse.kmi.open.ac.uk/open_dataset) [7] (https://doi.org/10.7910/DVN/GVLFXO) [8] (https://research.moodle.org) [9] (http://moocdata.cn/data/user-activity) [10] (https://github.com/riiid/ednet)

a selection of datasets (*e.g.*, [157, 161]), as presented in Table 3. These datasets have been widely employed in various educational research studies and have contributed significantly to the advancement of the field. Since some studies collect proprietary datasets that are not publicly accessible, this paper focuses exclusively on public datasets to support researchers in the field. Table 3 offers essential information about each dataset, including its name, URL, description, application scenarios, and literature applied.

We classify the collected datasets by source into three categories: Datasets Used for Competitions, Datasets from Online Education Platforms, and Open Data Repositories.

3.1.1 Datasets Used for Competitions

ASSISTments [120]. The ASSISTments dataset has many versions, such as Assistant 2009, 2012 and 2017, which have been used in several EDM competitions to promote research and development on educational technology and learning analytics. One of the most well-known competitions is ASSISTments Data Mining Competition.

KDD Cup 2010 [137]. This dataset is used in the KDD Cup 2010 EDM Challenge, which requires participants to use the student interaction logs contained in the provided dataset to train a new learning model and ultimately judge the results based on how accurately their model predicts student responses to new questions. This dataset is, therefore, also widely used for problems such as Knowledge Tracking.

3.1.2 Datasets from Online Education Platforms

Junyi Academy Online Learning Activity Dataset [119]. It is sourced from the Junyi Academy online learning platform, which provides personalized learning resources and support for students. The dataset consists of over 72,000 students' records of more than 16 million practice attempts on the platform within one year, from August 2018 to July 2019.

Canvas Network Dataset [150]. It is derived from the Canvas Network platform, which provides educational institutions and educators with the tools and resources to create and deliver online courses. The dataset contains a lot of course information as well as user interaction logs, etc.

Learn Moodle [156]. The dataset is derived from the Moodle Learning Management System, a widely used open source online learning platform that supports educational institutions and teachers in creating, managing and delivering online courses.

XuetangX [158]. The XuetangX dataset is derived from the online education platform XuetangX. It is a well-known online education platform in China, established in 2013, which provides MOOCs and other online learning resources, video viewing behavior, assignment submission and discussion participation, etc.

EdNet [160]. It is an extensive educational dataset amassed by Santa, an AI-based online teaching platform with a user base of 780,000 in South Korea. It encompasses two years' worth of student learning behavior data. It offers a plethora of information on student-system interaction, including knowledge tracking, cognitive processes, learning analysis data, and comprehensive records of students' online learning activities.

3.1.3 Open Data Repository

MOOCCube [126]. It is an open data repository for researchers in natural language processing, KG, and data mining related to large-scale online education, containing 706 real online courses, 38,181 instructional videos, 114,563 concepts, hundreds of thousands of course selections from 199,199 MOOC users, video viewing records, and a supplemental repository containing hundreds of thousands of academic paper resources related to in-class concepts. The concept description data is from Baidu and Wikipedia, and the course data and student behavior data are from XuetangX. Academic paper data is obtained from Aminer, a large-scale academic search engine.

xAPI-Educational Mining Dataset [145, 146]. This Dataset refers to a dataset based on the Experience API (xAPI) standard. xAPI is an open learning technology specification for recording learner behavior and interaction data in a variety of learning environments. The links in the table point to a publicly available xAPI-compliant database, Students' Academic Performance Dataset, stored on Kaggle, which contains 480 samples with sixteen features.

3.2 Processing Tools

There are many commonly used tools for processing and analyzing educational data in educational scenarios to provide powerful support and insight for educational researchers and teachers. This section will focus on several representative tools, including LOCO-Analyst, Datashop, SNAPP, GISMO, Meerkat-ED etc. These tools have unique functions and features in educational data processing that can help education practitioners better understand the learning process and optimize instructional design and practice. By using a combination of these tools, educational researchers and teachers can deeply analyze data on student learning

Table 4 Dataset processing tools information

Tool Name	Description
GISMO[1]	Graphical Interactive Student Monitoring Tool for Moodle (GISMO) provides an intuitive graphical interface to visually display information about student learning activities, engagement, grades, and progress
Meerkat-ED[2]	Meerkat-ED is an educational data analysis tool designed to help educators and researchers conduct in-depth analysis of student learning data. It generates comprehensive summaries of participants' engagement in discussion forums, illustrating their interactions, identifying discussion leaders and peripheral students, and providing various additional insights
Datashop [3]	DataShop is a collection of datasets and tools for educational mining. It not only collects and provides a large amount of educational data, including students' interaction data, response records, learning trajectories, assessment results, etc. in online learning environments, but also provides a series of tools and APIs for processing and analyzing these data, facilitating researchers to perform data mining, model building and evaluation
SNAPP[4]	Social Networks Adapting Pedagogical Practice (SNAPP) is a software tool that allows users to visualize the network of interactions generated by posts and responses in discussion forums. The network visualization of forum interactions will provide teachers with the opportunity to quickly identify patterns of user behavior
LOCO-Analyst[5]	LOCO-Analyst is an advanced educational tool designed to support teachers in evaluating and improving web-based learning environments. It provides valuable insights and feedback covering all aspects of the learning process, helping educators enhance the content and structure of their online courses and providing targeted feedback and recommendations so they can optimize course design and pedagogy
StREAM[6]	StREAM is a Student Engagement Analytics Platform, which is also a predictive algorithm developed by Solutionpath that includes educators with visualization of student engagement levels and identification of students who need help with certain tasks. For students, it provides information about the progress and status of their learning, enabling students and educators to adjust their learning or teaching strategies in a timely manner to improve teaching effectiveness

[1] (https://gismo.sourceforge.net)[2] (http://www.reirab.com/MeerkatED)[3] (https://pslcdatashop.web.cmu.edu/index.jsp)[4] (https://web.archive.org/web/20120321212021)[5] (http://jelenajovanovic.net/LOCO-Analyst)[6] (https://www.solutionpath.co.uk/stream)

behaviors, engagement, and learning outcomes, and gain valuable insights from them. The names, links, and detailed descriptions of these tools are provided in Table 4.

4 Future Directions

Deep Learning has shown great promise in EDM. Successful techniques in EDM based on Deep Learning can provide valuable insights to improve teaching, learning, and assessment. On the one hand, there are several potential insights into successful techniques for EDM using Deep Learning. On the other hand, we point out some practical challenges of Deep Learning in EDM and corresponding strategies to address them.

4.1 Learning Analysis and Intervention

Existing EDM methods often involve offline analysis. Future work could investigate the use of real-time learning algorithms, *e.g.*, deep reinforcement learning [162] or real-time recurrent learning algorithms [163], to offer timely insights and interventions for a more responsive educational environment. In addition, we can combine Multi-Task Learning [164] with Attention mechanisms to analyze student interaction data from learning management systems or other educational platforms, providing knowledge tracing and option tracing [165] into engagement and learning patterns. Multi-Task Learning can also be used to identify learners at risk of struggling academically and dropping out, thus allowing for early interventions and support.

4.2 Social Network Analysis and Collaboration

By learning directly from the graph structure and node features, GCNs can capture complex patterns in social networks, helping to analyze social networks within educational settings and revealing patterns of collaboration and communication among students. These insights can help educators design group activities and assignments more effectively or identify students who may benefit from additional support or social engagement opportunities. Besides, future work should encourage collaborations between computer scientists, educators, and psychologists. Although this direction does not focus on a specific algorithm, it emphasizes the importance of interdisciplinary knowledge in refining existing algorithms or developing new ones for EDM. For example, to make cross-domain recommendations for users in the educational environment, we can use the preference-aware Graph Attention Network [166], which leverages collaborative KG to capture user preferences within the domain and across the domain.

4.3 Large Language Models for Education

Large Language Models (LLMs) [167] are transforming many fields, including EDM. Their capacity to understand,

generate, and complete texts makes them a valuable tool in education. For instance, LLMs could be used to generate personalized educational content tailored to each learner's needs, interests, and proficiency level. To this end, we could focus on in-context learning or task-specific prompt methods for LLMs, such as GPT-4, to achieve better performance in such tasks. Besides, LLMs could help create advanced Intelligent Tutoring Systems (ITSs) [168] that can understand and respond to student queries in a contextually appropriate manner. Future work should integrate LLMs into existing ITSs and examine their impact on student learning outcomes.

It would be interesting that GPT-based architectures [169, 170] are employed to automatically score student essays or assess written responses, which can reduce the workload for educators and provide consistent evaluation. Such architectures are also competent to analyze and understand student language use, enabling educators to identify areas where students, even teachers, struggle with understanding or communication.

The LLM agent [171] represents a groundbreaking development in the realm of education technology. The LLM agent excels in generating high-quality educational content. Thus, learners can interact with the agent in a conversational manner, enabling a more personalized and adaptive learning experience tailored to individual needs. Moreover, the LLM agent promotes inclusivity in education by offering multilingual support. It can seamlessly communicate in various languages, breaking down language barriers and providing access to educational resources for a global audience.

4.4 Multimodal Learning Analytics

Many current EDM methods primarily rely on structured data. However, educational experiences produce a wealth of unstructured and semi-structured data (*e.g.*, image, audio, video, and even biometric data), which provide a more comprehensive understanding of student learning experiences. Deep Multimodal Learning algorithms [172], which combine CNNs for image/video data, RNNs or LSTMs for temporal data, and Transformers for textual data, could be used to leverage this wealth of information. These insights can inform the design of multimodal learning environments and interventions, accommodating diverse learning styles and preferences.

Based on the above representation of multimodal learning, multimodal affective computing and emotion recognition may be a promising line in EDM. To this end, we need well-designed models to analyze student facial expressions, speech, or physiological signals to infer emotional states and engagement during learning activities. In this way, they help educators adapt their teaching strategies to better meet students' emotional needs and improve overall learning experiences. For instance, hybrid contrastive learning [173] can be employed for multimodal sentiment analysis, in which semi-contrastive learning and intra-/inter-modal contrastive learning learn multiple relationships from cross-modal interactions.

4.5 Explainable Artificial Intelligence in EDM

Deep Learning techniques often yield models that are complex and difficult to interpret. Given the 'black box' problem associated with Deep Learning models, efforts should be made to create more transparent and interpretable models. Especially in EDM, making these models explainable and transparent becomes increasingly important since education usually places great emphasis on the scientific nature and causal relationships of things.

Future research could aim to develop or improve methods for generating understandable explanations of model predictions, *e.g.*, Deep Learning Important FeaTures DeepLIFT) [174] for the interpretable models used in student performance prediction, and Local Interpretable Model-Agnostic Explanations (LIME) [175] or KGs for explainable recommendation systems. Moreover, for applications that involve sequential data, such as studying student interaction with a learning management system over time, models like LSTM or GRU could be enhanced with explainability features. To this end, sequence interpretation methods, *e.g.*, Layer-wise Relevance Propagation (LRP) [176], can be used to explain sequential learning patterns.

4.6 Scalability and Real-time Processing

Educational data can be vast, especially in large institutions or online learning platforms, making it difficult to develop algorithms that scale efficiently. To tackle scalability issues, institutions can leverage scalable data processing frameworks such as Apache Hadoop and Spark. These frameworks can manage large datasets efficiently and are well-suited for the distributed nature of educational data. Cloud computing services offer flexibility and scalability, reducing the need for extensive on-premises hardware.

Real-time processing is imperative for adaptive learning systems, which must respond dynamically to student inputs. However, Deep Learning techniques usually fail to handle such tasks. This necessitates substantial computational resources and flexible mechanisms. We can employ stream processing frameworks, such as Apache Kafka [177], to enable the real-time analysis of data as it is generated, thus facilitating the development of adaptive learning environments. It is essential to invest in robust technological infrastructure and continually update it to keep pace with advancements.

In summary, while Deep Learning has already demonstrated its potential in EDM, there are quantities of exciting opportunities for further exploration and innovation. By focusing on the future directions highlighted above, we hope to promote significant progress in the field, contributing to the transformation of educational practices and the improvement of educational experiences.

5 Conclusion

Deep Learning algorithms have been widely applied in various fields. They have shown great potential in EDM to assist in improving the quality of modern education. In this survey, we first offer an extensive outline of the current state-of-the-art in Deep Learning-based EDM, highlighting three categories of Deep Learning (*i.e.*, unsupervised learning, supervised learning, and reinforcement learning) applied to four main educational scenarios. Additionally, we draw designs for knowledge tracing schema, student behavior detection schema, and personalized recommendation framework to demonstrate their principles intuitively. Secondly, a thorough overview of public datasets and processing tools for EDM is elaborated. Lastly, to provide new opportunities for innovation and improvement in this area, we put forward some promising future directions. This survey aims to inspire further research, collaboration, and progress toward broadening the application scope of EDM with Deep Learning.

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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